AIML Transaction & Fraud Analysis

Business insight report based on the supplied dashboard screenshots **Prepared by** Preethi

Dashboard overview

The dashboard combines transaction volume, transaction value, account count, opening balances and fraud flags. Charts cover transactions over time, alert types and the fraud / non-fraud split, with slicers for IS_FRAUD and ALERT_TYPE.

Headline figures: Transactions ~1.32M | Fraudulent transactions 1,719 (0.13%) | Accounts 10K | Total transaction amount 153.48bn | Average transaction 115.99K | Total initial balance 2.80M

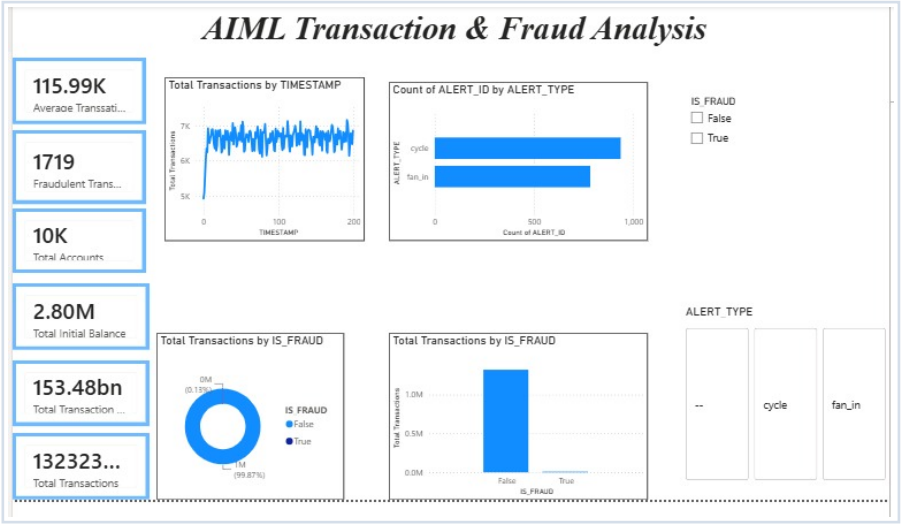
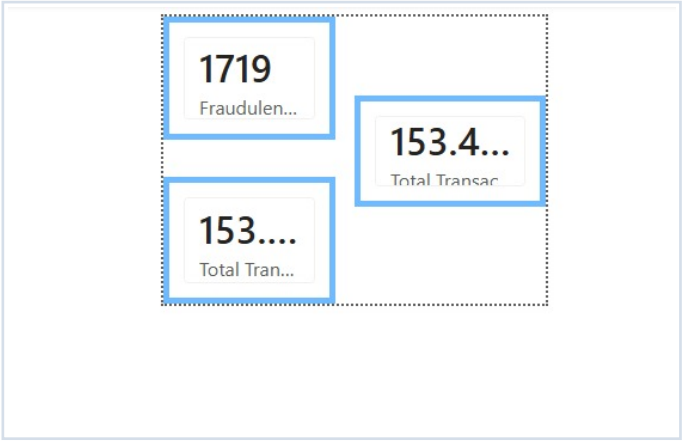


Figure 1. Supplied AIML Transaction & Fraud Analysis dashboard overview

1. KPI and fraud exposure analysis

Visible dashboard measures, with interpretation limited to the supplied screenshots

KPI	Visible value	Reading
Total transactions	132323… (≈1.32M)	Count of transactions; label is truncated
Fraudulent transactions	1719	Transactions flagged IS_FRAUD = True
Total accounts	10K	Accounts in the dataset
Total transaction amount	153.48bn	Sum of transaction values; currency not shown
Average transaction	115.99K	Average transaction amount
Total initial balance	2.80M	Sum of opening balances across accounts



The screenshots show roughly 1.32M transactions across 10K accounts, of which 1,719 are flagged as fraud. Two checks hold together: 153.48bn ÷ ~1.32M ≈ 116K, matching the average transaction of 115.99K, and 1,719 ÷ ~1.32M ≈ 0.13%, matching the donut chart.

Data caution

- Fraud *value* is not shown, only fraud *count*; do not estimate financial loss from it.
- Total transaction amount (153.48bn) is far larger than total initial balance (2.80M), which suggests simulated or synthetic data; confirm the source before drawing business conclusions.
- Currency is not displayed.

2. Transaction trend and alert mix

Transactions by timestamp and alert types shown in the supplied dashboard

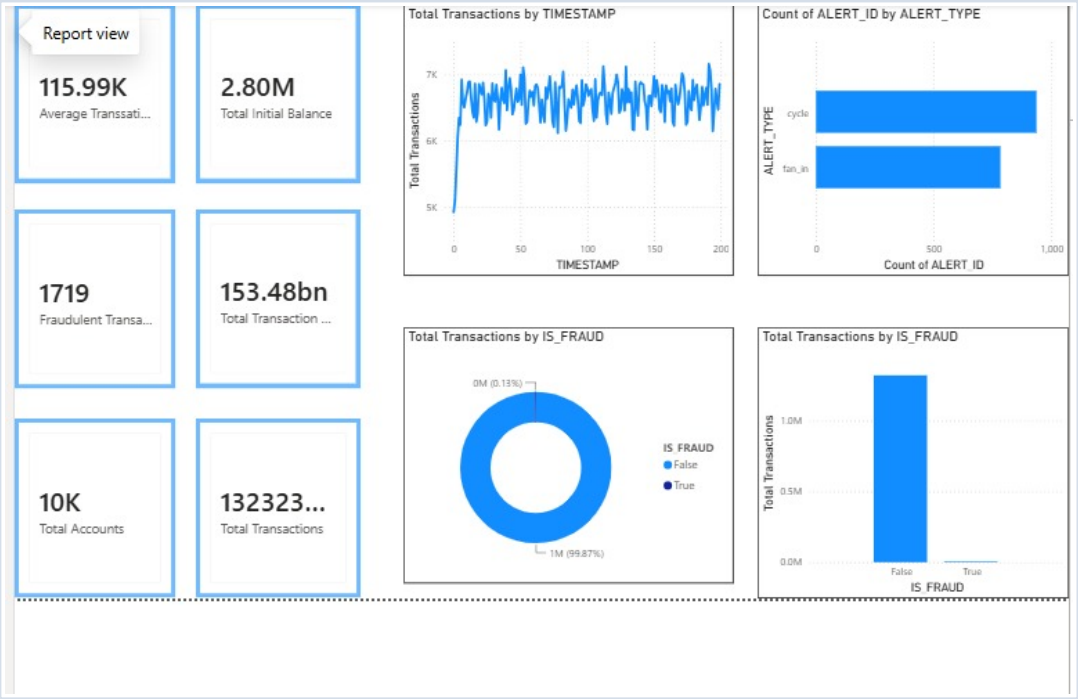


Figure 2. KPI cards, transaction trend, alert types and fraud split

Transaction trend

Transactions per timestamp start near 5K at timestamp 0, climb quickly to about 6.5K, then fluctuate between roughly 6K and 7K through timestamp 200 with no sustained rise or fall. The early ramp looks like a start-up effect. The y-axis starts at 5K, so the swings look larger than they are. This is a pattern observation only; it does not explain the cause.

Alert mix

Alerts are split between two types: **cycle** (about 950) and **fan_in** (about 780). Cycle alerts are the larger group. Together they are close to the 1,719 fraudulent transactions, which suggests the alert types describe the fraud patterns, but the dashboard does not confirm that link.

Chart values are visual estimates where labels are not printed.

3. Fraud profile and class balance

Fraud versus non-fraud split and what it means for monitoring and modelling

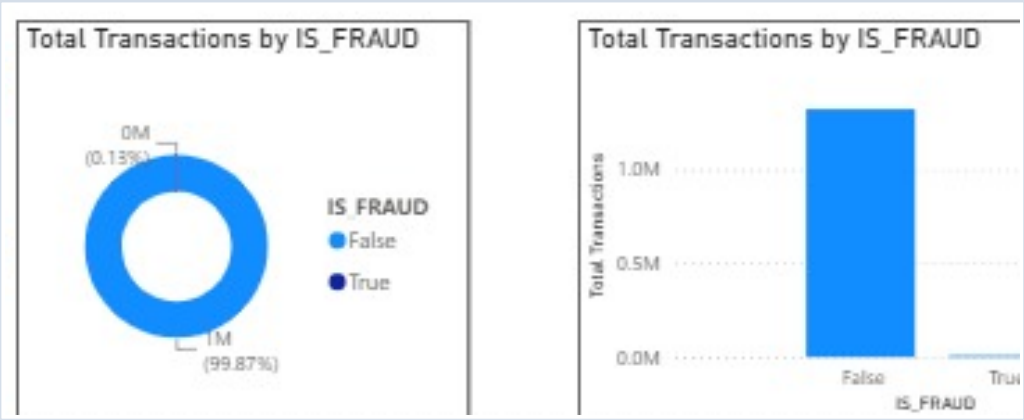


Figure 3. Total transactions by IS_FRAUD (donut and bar)

Fraud rate

Fraud makes up 0.13% of transactions (1,719 of ~1.32M); 99.87% are non-fraud. In the bar chart the fraud bar is barely visible next to the non-fraud bar.

Why imbalance matters

With so few fraud cases, a model that labels everything as non-fraud would still be about 99.87% accurate. Accuracy alone is therefore a poor measure; precision, recall and PR-AUC give a truer picture.

Slicer check

The ALERT_TYPE slicer shows a "--" option beside cycle and fan_in. This likely means no alert, but it is worth confirming that it is not a data-quality gap.

5. Business insights and recommendations

Actions grounded in the visible dashboard patterns

Observation	Recommendation
Fraud is 1,719 of ~1.32M transactions (0.13%), so fraud is rare but non-trivial in count.	Evaluate detection with precision, recall and PR-AUC. Use resampling or class weights if training a model.
Alerts are of two types, cycle (~950) and fan_in (~780), and together roughly match the fraud count.	Check whether each alert maps to a fraudulent transaction, then add detection rules for both patterns, starting with cycle.
Transaction volume is steady (about 6K to 7K per timestamp) after an initial ramp-up.	Treat the early ramp as warm-up, exclude it from baselines, and set alert thresholds on the stable range.
Total transaction amount (153.48bn) far exceeds total initial balance (2.80M), and the fraud amount is not shown.	Confirm the data source and add fraud amount, currency and account-level detail before judging financial impact.

6. Conclusion and data notes

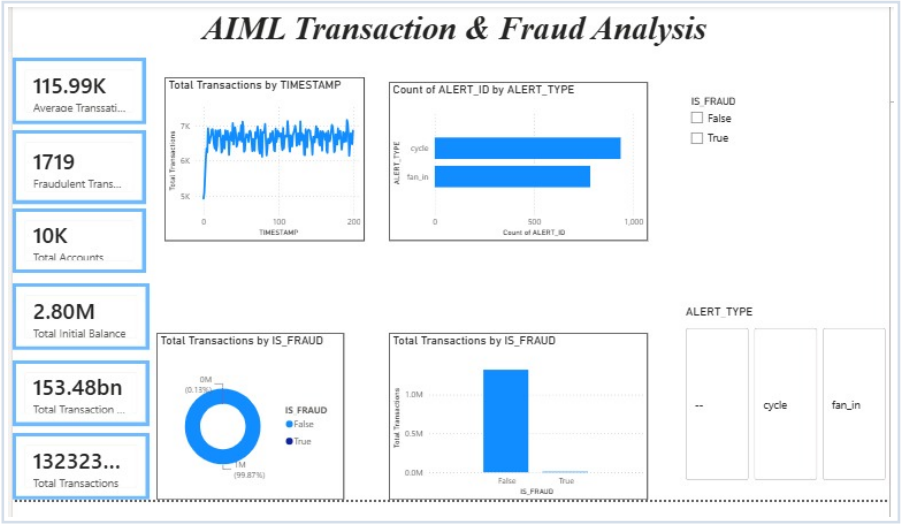
Summary of what the supplied dashboard supports

Conclusion

The dashboard gives a concise view of transaction activity and fraud exposure: about 1.32M transactions, 10K accounts, 1,719 fraud cases (0.13%) and two alert types. It supports volume monitoring, fraud-rate tracking and alert-mix review, but not conclusions about financial loss, fraud causes or model performance.

Data notes

• All figures come from the supplied screenshots; chart readings are visual estimates.
• Total transactions is truncated (132323…); ~1.32M is inferred from the fraud rate and average amount.
• Applied filter state for the IS_FRAUD and ALERT_TYPE slicers is not documented.
• No fraud amount, currency, customer, time-unit or model-performance data appears.



Source: supplied AIML Transaction & Fraud Analysis dashboard screenshots